

Land Survey

Engineering Mathematics I for Land Survey

Professional Learner Notes

Arithmetic, coordinates, bearings, trigonometric ratios and distance calculations for survey fieldwork.

Section	Purpose
Survey arithmetic	Units, rounding and field-book accuracy.
Coordinates	Eastings, northings and differences.
Bearings	Direction, quadrants and whole-circle bearings.
Trigonometry	Heights, offsets and inaccessible distances.
Distance	Coordinate distance and field measurement checks.

How to Study This Topic

The learning method follows a programmed engineering mathematics approach: understand the concept, study the formula, follow a worked method, then practise with a technical problem. The aim is not memorisation only; the learner must interpret the answer in the department context.

- Write the formula before substitution.
- Use correct units.
- Check whether the result is reasonable.
- State a technical interpretation.
- Repeat practice until the method is fluent.

Formula Summary

Distance from coordinates

$$d = \sqrt{(\Delta E)^2 + (\Delta N)^2}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Bearing angle

$$\theta = \tan^{-1} \left(\frac{\Delta E}{\Delta N} \right)$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Sine

$$\sin \theta = \frac{O}{H}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Cosine

$$\cos \theta = \frac{A}{H}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Tangent

$$\tan \theta = \frac{O}{A}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

1. Survey arithmetic

Units, rounding and field-book accuracy.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$d = \sqrt{(\Delta E)^2 + (\Delta N)^2}$$

Worked method

Example: Coordinate distance. A point changes 60 m east and 80 m north. Find distance.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports field distances. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving heighting. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving field distances. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving coordinate differences. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving heighting. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.

2. Coordinates

Eastings, northings and differences.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\theta = \tan^{-1} \left(\frac{\Delta E}{\Delta N} \right)$$

Worked method

Example: Bearing. Find the bearing from coordinate differences $\Delta E = 45$ m and $\Delta N = 75$ m.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports coordinate differences. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving heighting. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving field distances. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving coordinate differences. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving heighting. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving field distances. Show formula, substitution, units and interpretation.

3. Bearings

Direction, quadrants and whole-circle bearings.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\sin\theta = \frac{O}{H}$$

Worked method

Example: Heighting. Use angle of elevation and horizontal distance to estimate height.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports bearings. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving field distances. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving coordinate differences. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving heighting. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving field distances. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving coordinate differences. Show formula, substitution, units and interpretation.

4. Trigonometry

Heights, offsets and inaccessible distances.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\cos\theta = \frac{A}{H}$$

Worked method

Example: Coordinate distance. A point changes 60 m east and 80 m north. Find distance.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports heighting. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving field distances. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving coordinate differences. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving heighting. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving field distances. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving coordinate differences. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.

5. Distance

Coordinate distance and field measurement checks.

In land survey, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\tan\theta = \frac{O}{A}$$

Worked method

Example: Bearing. Find the bearing from coordinate differences $\Delta E = 45$ m and $\Delta N = 75$ m.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports control points. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving coordinate differences. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.
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4. Create and solve one technical calculation involving control points. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving field distances. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving coordinate differences. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving heighting. Show formula, substitution, units and interpretation.

Mixed Revision Questions

1. A land survey problem involves coordinate differences. Choose a suitable mathematical method, solve the problem, and explain the result.
2. A land survey problem involves bearings. Choose a suitable mathematical method, solve the problem, and explain the result.
3. A land survey problem involves heighting. Choose a suitable mathematical method, solve the problem, and explain the result.
4. A land survey problem involves control points. Choose a suitable mathematical method, solve the problem, and explain the result.
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9. A land survey problem involves control points. Choose a suitable mathematical method, solve the problem, and explain the result.
10. A land survey problem involves field distances. Choose a suitable mathematical method, solve the problem, and explain the result.
11. A land survey problem involves coordinate differences. Choose a suitable mathematical method, solve the problem, and explain the result.
12. A land survey problem involves bearings. Choose a suitable mathematical method, solve the problem, and explain the result.
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14. A land survey problem involves control points. Choose a suitable mathematical method, solve the problem, and explain the result.
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17. A land survey problem involves bearings. Choose a suitable mathematical method, solve the problem, and explain the result.
18. A land survey problem involves heighting. Choose a suitable mathematical method, solve the problem, and explain the result.
19. A land survey problem involves control points. Choose a suitable mathematical method, solve the problem, and explain the result.
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21. A land survey problem involves coordinate differences. Choose a suitable mathematical method, solve the problem, and explain the result.

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23. A land survey problem involves heighting. Choose a suitable mathematical method, solve the problem, and explain the result.
24. A land survey problem involves control points. Choose a suitable mathematical method, solve the problem, and explain the result.
25. A land survey problem involves field distances. Choose a suitable mathematical method, solve the problem, and explain the result.
26. A land survey problem involves coordinate differences. Choose a suitable mathematical method, solve the problem, and explain the result.
27. A land survey problem involves bearings. Choose a suitable mathematical method, solve the problem, and explain the result.
28. A land survey problem involves heighting. Choose a suitable mathematical method, solve the problem, and explain the result.
29. A land survey problem involves control points. Choose a suitable mathematical method, solve the problem, and explain the result.
30. A land survey problem involves field distances. Choose a suitable mathematical method, solve the problem, and explain the result.

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.
2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.
3. Uploaded KNEC solved mathematics materials for examination-style orientation.
4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.